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Accurate measurements of laminar burning velocity using the Heat Flux method and thermographic phosphor technique

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Abstract

The Heat Flux method was further developed to significantly reduce its experimental uncertainty and used to determine burning velocities under conditions when the net heat loss from the flame to the burner is zero. Non-stretched flames were stabilized on a perforated plate burner at 1 atm. Measurements of the adiabatic burning velocity of methane/air flames at initial mixture temperatures of 318 K are presented. Previously, tiny thermocouples in the thin burner plate were used to evaluate when the heat flux of the flame to the burner is zero. Related errors limit the accuracy of the method so far. A new experimental procedure based on thermographic phosphors is described which avoids these errors. The new experimental procedure is described. An UV thermographic phosphor ZnO:Zn was selected and used to sensitively control the temperature uniformity on the burner plate to within 60 mK. Uncertainties of the measurements were analyzed and assessed experimentally. A more accurate evaluation of the gas velocities, using mass weighting, was introduced to increase the accuracy further. The uncertainty of the measured adiabatic burning velocities due to the temperature scattering can be reduced from typically ± 1.5 cm/s at $\Phi = 0.7$ methane/air flame to ± 0.25 cm/s. The overall accuracy of the burning velocities including the uncertainty from the employed mass flow controllers was evaluated to be better than ± 0.35 cm/s. Possibilities to further improve the measurement accuracy are discussed with practical considerations.
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Keywords: Burning velocity; Heat Flux method; Thermographic phosphor

1. Introduction

The adiabatic laminar burning velocity is a fundamental parameter of each combustible mixture, which depends on the stoichiometric ratio, pressure and temperature. The laminar burning velocity at standard conditions, i.e. atmospheric pressure and initial temperature of 298 K is invaluable for the characterization of combustion properties of the given fuel, for understanding of the underlying chemistry, validation of models, etc.

Comparison of kinetic mechanisms often serves to prove that one or another model is the best in predicting new experimental data. Comparison and validation of the models is usually based on visual perception; this qualitative

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judgment could be acceptable only when the disparity between measurements and predictions of different mechanisms is obvious and exceeds experimental uncertainties. Model development and improvement, however, requires a quantitative measure of the discrepancy and evaluation of the sensitivity of the model predictions to variation of reaction rates or thermo-chemistry. An example of such a quantitative analysis was given by Brown et al. [1], who demonstrated that variation of each rate constant within its evaluated uncertainty results in the uncertainty in the computed burning velocity of about 8–10% for hydrogen flames and 12–13% for methane flames. This means that the accuracy of the measured laminar burning velocities should be below these values to be useful for the model development.

Burning velocities of many different fuels as a function of temperature, pressure, and stoichiometric ratio have been measured by many investigators; however, the data were incompatible for a long time. Wu and Law [2] demonstrated experimentally that the flame stretch due to flame front curvature and/or flow divergence must be taken into account in the data processing. The methodology of the determination of laminar burning velocities with the counter-flow twin-flame technique by extrapolation to zero stretch rate has been further improved by Vagelopoulos et al. [3]. The uncertainty of the counter-flow twin-flame technique is limited by the accuracy of the LDV (laser Doppler velocimetry) technique, which was estimated to be 1–2 cm/s for burning velocities less than 60 cm/s and up to 10 cm/s for those above 170 cm/s [4].

In constant-volume bombs, the un-stretched burning velocities were determined in many research groups using essentially the same approach. The experimental uncertainty of the laminar burning velocities derived from expanding spherical flames was estimated [5,6] to be about 10%. Recent work aimed at improving methodology [7] demonstrates a maximum random error of 1.45% for the observed flame speed; the uncertainty of the derived laminar burning velocity, however, may reach up to 2 cm/s in rich mixtures.

The Heat Flux method for the stabilization of adiabatic premixed laminar flames on a flat flame burner has been proposed by de Goey et al. [8] and further developed by van Maaren et al. [9,10]. This method was extensively used for measuring laminar burning velocities of gaseous fuels, e.g. [9–12]. Detailed analysis of the uncertainties of this method was performed [11,13] and showed that the overall accuracy of the burning velocity measurements is better than ± 1 cm/s (double standard deviation with 95% confidence level). The relative accuracy of the equivalence ratio was found to be below 1.5%.

One can conclude that contemporary experimental methods for measuring laminar burning velocities have accuracy of the order of 1–2 cm/s and for the purpose of model validation and development the accuracy should be improved. This is particularly important in slowly burning lean and rich flames; burning velocities of hydrocarbons mixed with air are of the order of 10–20 cm/s at these conditions, and thus the uncertainty is here up to 10%, which is relatively too high. The present work therefore aimed at significant reduction of the experimental uncertainty of the Heat Flux method. Two major sources of experimental uncertainties pertinent to the Heat Flux method were identified [11,13] as follows: irregular thermocouple placement in the burner plate and inaccuracy in the mass flow control. The series of thermocouples attached to the burner plate are essential in the measurements of the temperature distribution as will be described below. In the present work an alternative technique to measure the surface temperature using thermographic phosphors was successfully employed.

Thermographic phosphors (TP) are ceramic crystal substances often in the form of a fine powder. Usually they are doped with some kind of rare earth metal, which gives them specific phosphorescence properties. These phosphorescence properties are sometimes temperature dependent, which makes TP a sensor in temperature measurements applications. The use of TP is well established for measuring temperature on surfaces [14–18]. By applying laser-induced phosphorescence (LIP), temperature can be monitored by registering the change in either phosphorescence lifetime or the change in spectral profile. Both point- and two-dimensional measurements are possible. The point measurement approach usually utilizes the temperature dependence of the phosphorescence signal lifetime to retrieve temperature information [19]. Exploration of the temperature dependence of the spectral profile makes it possible, with the use of suitable optical filters, to do 2D temperature measurements on a surface, either through the use of two separate detectors or a single detector equipped with an image-doubling device (stereoscope) [20,21]. With suitable choice of phosphor material, measurements can be carried out at temperatures from cryogenic values to 2000 K. Due to interference with background Planck radiation at higher temperature, phosphors with emission in the blue spectral region are preferable in order to increase the signal-to-noise ratio. In addition short lifetime of the phosphorescence gains the possibility to time gate for background discrimination [22].

In the following the principles of the Heat Flux method are outlined. Then the choice of the suitable thermographic phosphor and optical arrangement are described. Finally, typical measurements are presented and compared with those using the classical configuration of the Heat Flux method. Uncertainties of the measurements were analyzed and assessed experimentally.

2. Experimental details

2.1. Heat flux burner facility

A detailed description of the Heat Flux method and associated experimental uncertainties are given elsewhere [11,13]. Important features of the method are, therefore, only shortly outlined in the following. This paper reports new developments to significantly reduce its experimental uncertainty. The key part of the experimental setup for the adiabatic flame stabilization using the Heat Flux method is the burner shown in Fig. 1. A 2 mm thick burner plate perforated with small holes (0.5 mm in diameter) is attached to the burner outlet. The burner head has a heating jacket supplied with thermostatic water to keep the temperature of the burner plate constant. During the experiments this temperature was fixed at 358 K. The plenum chamber has a separate cooling system supplied with water at a temperature of 298 K (standard conditions), or could be increased up to 358 K. This fixes the initial temperature of the fresh gas mixture. The heating jacket keeps the burner plate edges at a certain temperature higher than the initial gas temperature, thus warming up the (unburned) gases flowing through. Conductive heat transfer of the flame to the burner plate cools the gas flow on its turn. If the flame is stabilized under sub-adiabatic conditions, the gas velocity is lower than the adiabatic flame burning velocity and the sum of the heat loss and heat gain is higher than zero. Then, the centre of the burner plate is hotter than the heating jacket. If the unburned gas velocity is higher than the adiabatic burning velocity (super-adiabatic conditions), the net heat flux is lower than zero and the centre of the burner plate is cooler than the heating jacket. By changing the flow rate of the gas mixture an appropriate value of the gas velocity can be found to nullify the net heat flux. In this case the radial temperature distribution in the burner plate is uniform and equal to the temperature of the heating jacket.

In the classical configuration of the Heat Flux method, a series of thermocouples attached to the burner plate as shown in Fig. 1 allows the temperature distribution measurements in it. The flow rate at which the net heat flux is zero is shown to be the adiabatic flame burning velocity [8,9].

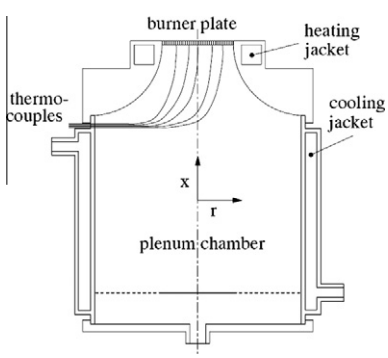

Fig. 1. The schematic of the heat flux burner.

A mixing panel was assembled for the gas supply from gas cylinders to the inlet of the plenum chamber of the burner. It consists of three ducts for the fuel, oxidizer and diluent gases. Each duct connected to its own gas cylinder has a buffer vessel and a Bronkhorst Hi-Tech mass flow controller (MFC). The 2 l buffer vessels are dimensioned according to the recommendation of the MFC manufacture. In the present work the MFCs were calibrated just before the experiments by measuring the mass change.

2.2. Thermographic phosphor

The temperature of the heat flux burner plate used in this work is around 358 K, so we need to choose an appropriate TP accordingly. There are several TP available in this range, among which zinc oxides are probably the best candidates, since they feature a very high temperature sensitivity in this temperature range. Zinc oxides have since long been utilized for a variety of applications such as UV-sun-blocking and LEDs [23]. The substance used in the experiment is ZnO:Zn with a particle size of 5 μm delivered by Phosphor Technology, UK.

ZnO:Zn has been investigated in detail by Särner et al. [24]. It is a blue emitting phosphor which, when excited with 355 nm laser, emits radiation with a peak at 388 nm. This UV/blue emission has a lifetime shorter than 1 ns, and is strongly red shifted with temperature. This phosphor also has a broad emission peak around 510 nm with a much longer lifetime of about 1 μs. With short camera exposure time, however, interference from this emission is avoided during spectral ratio temperature measurements.

The UV emission line of ZnO:Zn is according to Zhang et al. [25] originating from near-band-edge emission due to excitation states. The unusually large spectral line shift of ZnO:Zn enables temperature measurements with a very high precision from room temperature to 900 K, using the spectral ratio method. The filters used in this work are an interference filter at 390 ± 5 nm and a GG420 long pass filter. The ratio of the intensities through the two filters shows high temperature sensitivity.

2.3. Optical setup

The optical arrangement of the experimental setup is shown in Fig. 2. A thin layer of the ZnO:Zn was coated on top of the cap of each thermocouple, which can be seen in Fig. 3 as white dots labeled from one to eight. A Brilliant B laser system with a 3rd harmonic package was adopted to generate the 355 nm laser as light source. The laser was

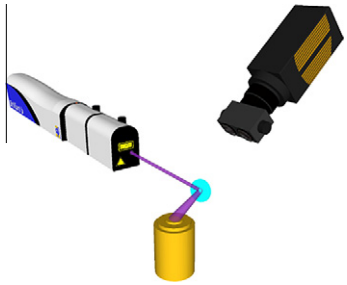

Fig. 2. The schematic of the experimental setup for the laser-induced phosphorescence measurements.

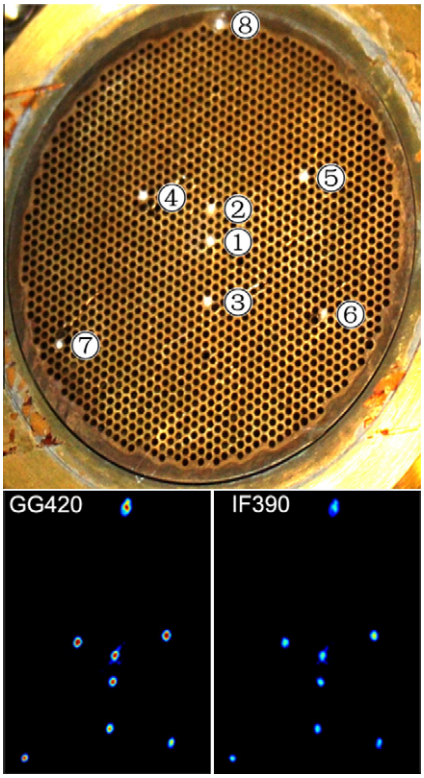

Fig. 3. Photograph of the perforated burner disc coated with eight thermographic phosphor dots on top of the installed thermocouples installed. Showing beneath are a typical twin laser-induced phosphorescence images collected through two different filters as marked in the corresponding images.

reflected and expanded to the burner disc by a concave mirror, which covered all the phosphor dots. The phosphorescence from the dots was then collected by a camera system, which includes a stereoscope, an objective (Nikon, 105 mm, f/2.5), and an ICCD camera (Princeton PI-MAX, 1024 × 1024 pixels). The stereoscope is an image-doubling device attached to the objective of the camera for

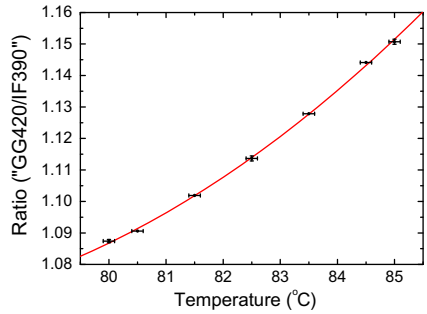

Fig. 4. Temperature sensitivity calibration curve of ZnO:Zn using the ratio between phosphorescence intensities collected with an interference filter (390 ± 5 nm) and a long pass filter (GG420).

simultaneous recordings of two optically identical images on a single CCD-chip [21]. An interference filter (390 ± 5 nm) and a long pass filter (GG420) were used in the stereoscope to obtain the two images on the CCD-chip. The ICCD camera was synchronized with the laser, and the gate width was set to be 30 ns. The typical twin phosphor images are shown in Fig. 3 as well. By calculating the ratio for the intensities between the two corresponding phosphor dots in the twin images, we can get the temperature distribution on the burner plate.

3. Results and discussion

3.1. Typical measurement

The sensitivity of the TP method has been tested on the burner plate by varying the temperature of the heating jacket water. The results are shown in Fig. 4, in which the error bars in horizontal direction represent temperature uncertainty of the water heating system (provided by the manufacture) and error bars in vertical direction represent the standard deviation of three independent measurements of the intensity ratio, which have been performed for temperatures at 80, 82.5 and 85 °C. For such an error bar, the corresponding uncertainty of the temperature is about 60 mK, which is in good agreement with the results of Särner et al. [24]. Hence the ratio from the twin images (see Fig. 3) can be converted into temperature and the temperature of eight dots can be obtained at each flame condition.

According to Bosschaart and de Goey [13], the temperature distribution along the radius of the burner disc can be written as:

$$T(r) = T_c + ar^2, \tag{1}$$

where T_c is the temperature in the burner disc centre, and a is the parabolic coefficient varied with the gas flow speed as described above. T_c and a

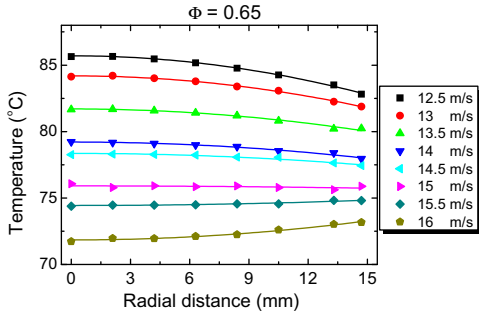

Fig. 5. Temperatures measured with thermographic phosphors at different radial distance on top of the burner disc (symbol) at different gas supply speeds and fitting curves (solid lines) for methane/air ($\Phi = 0.65$) flame speed evaluation.

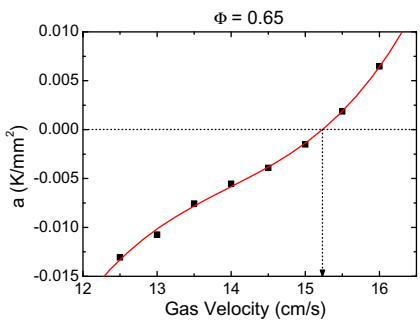

Fig. 6. The gas supply velocity dependence of the parabolic coefficient a, fitted with a third order polynomial function.

can be obtained by fitting the temperatures of the distributed eight dots with a parabolic curve. Interpolation of the relation curve between a and gas supply speed gives the adiabatic flame speed S_L with the Heat Flux method.

Figure 5 is an example of the temperatures measured on the eight TP coated dots (shown in Fig. 3) on top of the burner plate when burning a $\Phi = 0.65$ methane/air flame. The measured temperatures of the eight dots are represented by symbols and the parabolic fittings are represented by solid lines. Each fitted curve will give a parabolic coefficient as expressed in Eq. (1). Figure 6 is the relation curve between the parabolic coefficients a and the gas flow speeds fitted by the third order polynomial function. The flame speed can be determined as ~15.2 cm/s when the coefficient a approaches zero.

The described TP method was applied to the adiabatic laminar burning speed measurement of methane/air flames of a wide range of stoichiometric ratios (0.65–1.45) at initial gas mixture temperature of 318 K. The results are illustrated in Fig. 7 together with the results obtained with conventional thermocouple (TC) method for comparison. The two series show good agreement with a difference less than 0.5 cm/s, except for the very lean cases ($\Phi = 0.65$ and 0.7) where the difference is around 1 cm/s but still within the uncertainty of the TC method of about 2 cm/s [13].

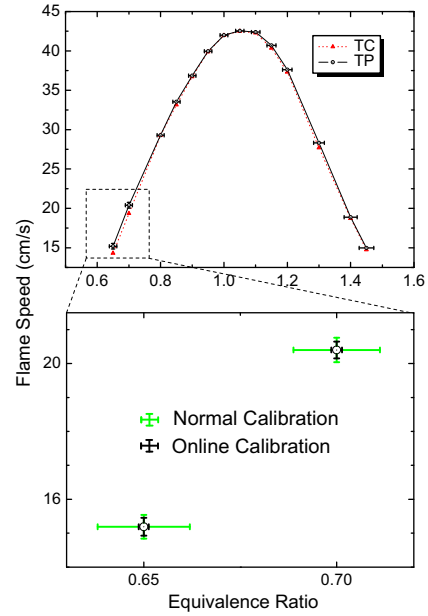

Fig. 7. Above, comparison of results of flame speed measurements obtained with thermocouple (TC) method and thermographic phosphor (TP) method. Below, error bars for the flame burning speeds measured with TP method for methane/air flames of $\Phi = 0.65$ and 0.7, black bars represent the possible values employing the online calibration method.

3.2. Error evaluation

The uncertainty of the adiabatic flame speed measured with the Heat Flux method originates mostly from two sources: the scattering of the measured temperatures from the parabolic fitting and the uncertainty of the employed MFCs.

3.2.1. Error from the temperature scatterings

In the measurement with the phosphor technique demonstrated in Fig. 5, the averaged temperature scattering is 50 mK and the maximum scattering is 160 mK, which is much better than thermocouple (typically 500 mK). The resulting error bar in the measured burning velocity can be estimated by the following equation [13]:

$$\sigma_{S_L} = \frac{1}{s}\frac{2\sigma_{tp}}{r_b^2}, \tag{2}$$

where s is a sensitivity parameter defined as:

$$s = \left.\frac{da}{dS_L}\right|_{a=0} \tag{3}$$

and can be extracted from Fig. 6; and r_b is the radial distance of the outermost TP dot which is 14.7 mm in this case; and σ_{tp} is the maximum temperature scatting 0.16 K in this measurement. The error bar of the measured flame speed σ_{S_L} was calculated to be 0.25 cm/s.

This measurement was performed for two lean flames with $\Phi = 0.65$ and 0.7, where the thermocouple method gives the largest error bars. Compared with the flame speed measurement uncertainties summarized in Ref. [11], the present measurement provides much higher precisions.

3.2.2. Error from the MFC

Due to the dramatic suppression of errors introduced by temperature scattering, the uncertainty aroused from MFC become dominant. Hence a proper calibration of the applied MFCs is crucial. In our laboratory, a convenient and highly precise MFC calibration system was built based on the precise measurement weight change. In the system, a bottle of sample gas connecting to the MFC with soft gas lines is placed on a platform scales. When the gas flow through the MFC is stable, the time on a stopwatch together with the value of the scales can be recorded simultaneously by taking an image of both displays. After a certain amount of gas flowing out, the time and the bottle weight are recorded again in the same way. Then the weight change per unit time will give the real mass flow rate of the MFC at the chosen opening percentage. This calibration method is superior to the commonly applied method in monitoring the volume change, which might be influenced by the ambient temperature and pressures.

During this calibration procedure; after several set-points (e.g. from 10% to 100%) being calibrated a fitting to a third order polynomial function is used to correct for non-linear effects. In this way, following the instructions given by the MFC manufacturer, the uncertainty (ΔF, ln/min) of the MFC (in the range of 10–100%) can be written as

$$\Delta F = 0.5\% \cdot \mathrm{Rd} + 0.1\% \cdot \mathrm{FS}, \tag{4}$$

where Rd is the reading of the MFC, and FS is the full scale of the MFC. Since the full scale of the MFC is involved in the error, over-dimension of the MFCs has to be avoided.

The uncertainty of the MFC will contribute to the errors for both burning velocity and stoichiometric ratio, which can be estimated by partial derivatives:

$$\frac{\Delta\Phi}{\Phi} = \frac{\Delta F_{\mathrm{CH_4}}}{F_{\mathrm{CH_4}}} + \frac{\Delta F_{air}}{F_{air}}, \tag{5}$$

$$\Delta S_L = \frac{\Delta F_{\mathrm{CH_4}} + \Delta F_{air}}{600 \cdot A}. \tag{6}$$

The uncertainties in stoichiometric ratio estimated by Eq. (5) are shown in Fig. 7 (horizontal error bar). It is ~0.01 when $\Phi < 1$, and it becomes ~0.02 when $\Phi = 1.45$. According to Eq. (6), where A is area of the burner plate in m^2 and ΔS_L in cm/s, the uncertainties from MFCs will add ~0.1 cm/s error to S_L when $\Phi = 0.65$ (and $\Phi = 1.45$), and it will add ~0.2 cm/s when $\Phi = 1$. In this work, the overall errors for two lean cases ($\Phi = 0.65$ and 0.7) were calculated and displayed in Fig. 7, marked as normal calibration. In this work, the error from the MFCs is of the same order as the error from temperature fitting scatters, which is not negligible anymore. Moreover, it introduces significant errors to the stoichiometric ratios. Further efforts are needed to decrease the errors from MFCs.

Here we propose an online calibration method (calibration during experiments), instead of the normal one (calibration before experiments). The weight changes of fuel and air bottles can be measured along with the experiments. Then the error from the MFCs will be determined only by the uncertainty of the scales (ΔS) and weight change (ΔM):

$$\Delta F = \frac{\Delta S}{\Delta M} \cdot \mathrm{Rd}. \tag{7}$$

Assuming that we have scales with precision $\Delta S = 0.01$ g, and for each online calibration $\Delta M \geqslant 10$ g, then Eq. (7) will become

$$\Delta F = 0.1\% \cdot \mathrm{Rd}. \tag{8}$$

According to Eqs. (5), (6) and (8), errors for both stoichiometric ratios and burning velocities based on online calibration were calculated. It has a value of less than 0.003 for all the stoichiometric ratios and adds less than 0.004 cm/s to the uncertainty of S_L for all the stoichiometric ratios. The overall error from normal calibration and online calibration are compared in Fig. 7, and the advantage of the online calibration is obvious.

3.2.3. Other resources of errors

One should remember that significant suppression of the errors caused by irregular thermocouple placement in the burner plate and the inaccuracy in the mass flow control probably brings the importance of other non-idealities of the Heat Flux method into view. A number of assumptions behind this method of the adiabatic flame stabilization have been validated numerically and experimentally employing Laser Doppler Velocimetry, Laser-Induced Fluorescence and Coherent Anti-Stokes Raman Scattering. The velocity profiles of the cold flow issuing from the burner mouth were found uniform with small distortions next to the burner edges [26,27]. The velocity profiles of the burnt gas above the flame were found to be symmetric and uniform in a core region of the flame. The measured maximum velocity variations in flames close to the adiabatic value were within 1–2% at radii smaller than about 80% of the burner radius [26,27]. This flow non-uniformity is compa-

rable to the uncertainty of the method achieved in the present work. Burners with a larger diameter might be required to suppress the impact of the edge-effects.

The small-scale structure of the flat flames stabilized on the perforated plate was analyzed by de Goey et al. [28] to reveal the range of flame velocities where the flame front is one-dimensional. It was concluded that with the drilling pattern of the burner plate employed here, the small flamelets converge completely forming a perfect 1-D flame at flow velocities below about 0.5 m/s. The criteria used for this evaluation are also comparable to the uncertainty of the method achieved. Further improvement may be gained in smaller diameter the holes and a reduction of the plate thickness.

It is believed that radicals do not touch the burner because of the presence of methane, which consumes the radicals very rapidly [29]. This chemical aspect of the method is difficult to evaluate experimentally as compared to the physical parameters which are under control. Although no indication of the influence of surface reactions was found by many investigations using this method, further experimental prove is required, which is the objective of the authors.

Radiative effects have a negligible effect (<0.2 cm/s) on the relevant heat fluxes in the system. Absorption of gas radiation by the burner and radiative heat losses by the burner itself has an error of the order of 2% on the relevant heat fluxes. This has been validated experimentally by changing the plate temperature, see [30].

4. Conclusions

Accurate measurements of laminar burning velocity have been demonstrated with Heat Flux method, which determine burning velocities of non-stretched flames stabilized on a perforated plate burner at 1 atm under conditions when the net heat loss from the flame to the burner is zero. Previously, tiny thermocouples mounted on several places in the thin burner plate served to evaluate the heat flux of the flame to the burner. Systematic errors of the order of 1 K in the measurements account for the bottleneck in the accuracy of the method so far. A new experimental procedure is described which avoids these errors. A UV thermographic phosphor ZnO:Zn was selected and used to monitor the temperature uniformity on the burner plate, which was proved to provide a sensitivity better than 60 mK for in situ measurements. Measurements of the adiabatic burning velocity of methane/air flames at initial mixture temperatures of 318 K are presented with the new experimental procedure and the measurements uncertainties were analyzed and assessed experimentally. The uncertainty of the measured adiabatic burning velocities due to the temperature scattering was reduced from typically ± 1.5 cm/s at $\Phi = 0.7$ methane/air flame to ± 0.25 cm/s. The overall accuracy of the burning velocities including the uncertainty from the employed mass flow controllers (MFCs) was evaluated to be better than ± 0.35 cm/s. A novel method for on line MFC calibrations was proposed, which in principle can be used not only to remove the uncertainties introduced by the MFCs but also to reduce substantially the error bars in the equivalent ratio values. Possible error bars from other resource was also discussed.

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